

Final Design Report  
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Section B: Wednesday

## Table of Contents

|                      |    |
|----------------------|----|
| Executive Summary    | 2  |
| Introduction         | 2  |
| Device Analysis      | 3  |
| Defining the Problem | 4  |
| Solution             | 5  |
| Prototype            | 7  |
| Conclusion           | 8  |
| References           | 10 |
| Team Summary         | 12 |
| Appendix             | 13 |

## Executive Summary

Tegaderm clear wound dressing is an effective alternative for traditional dressings as it enables healthcare professionals to prioritize patient care and closely monitor wound sites without disturbing the injury by repeatedly removing the dressing. Although Tegaderm dressing offers numerous benefits, the device's design is still flawed, particularly in terms of the adhesive used. The dressing's adhesive is primarily acrylate, which creates a firm seal against the skin, allowing the wound site to retain natural moisture and accelerate rate of epithelial regeneration (Nuutila, 2021). However, the peel strength required to break this seal and remove the dressing can cause undesired irritation in sensitive skin types; a prominent group affected by this design flaw is individuals with eczema. The removal process of the dressing can damage the outermost layer of the epidermis, leading to further eczema flare-ups and slower healing of the wound (Matsumura et. al., 2012).

The chosen design rationale to improve the Tegaderm's incompatibility with eczema-prone skin proposes to replace the acrylate with an equally economical silicone-based adhesive. This design change reduces the original peel strength—approximated to be equal to that of long-wear Band-Aids—of 250 g/cm<sup>2</sup> down to 50-100g/cm<sup>2</sup> (Sharma, 2024). The reduced peel strength enables users to remove the dressing without exerting excess force and causing subsequent irritation to the skin. Additionally, this change does not negatively affect the key traits of Tegaderm dressing, such as moisture retention or the clear appearance, allowing the improved design to fulfill its original purpose of accelerating wound healing while allowing healthcare professionals to monitor wound progression. When testing the two types of adhesives, our team focused on determining which material would remove less tissue from the epidermis. To identify the better alternative, equally sized acrylate and silicone adhesives were attached to sheets of corrugated cardboard, which served as a replication of human skin. Next, both adhesive materials were removed from their respective piece of cardboard using an equal force. The mass before and after each adhesive-cardboard system was recorded and interpreted as the mass of cardboard removed. If translated into a real-world setting, this quantity would be equated to the mass of epidermis removed from the skin. The testing revealed the acrylate adhesive removed 0.021 grams of cardboard while the silicone adhesive removed only 0.007 grams. Given the result, we determined the silicone adhesive would be an effective alternative to the acrylate material due to its implicated compatibility with sensitive skin. Replacing the original dressing's acrylate adhesive with a silicone-based alternative can improve Tegaderm's compatibility with a greater range of users and ultimately improve patient care on a global scale.

## Introduction

The Tegaderm clear wound dressing revolutionizes wound care by allowing medical professionals to actively observe the progress of healing without removing the dressing and constantly disrupting the wound site. The dressing can be applied to protect open wounds, but also to secure IV lines, catheters, and other medical devices while keeping the covered skin hydrated (Britto et. al., 2024). Additionally, the flexibility of the product allows the dressing to create a membrane-like cover, minimizing any disruption to regular movement. The dressing is composed of an acrylate adhesive paired with a polyurethane film, which, although it offers the benefit of being hypoallergenic, can be excessively harsh on sensitive skin with conditions such as eczema (3M, n.d.). While the acrylate adhesive may not directly induce a reaction on the covered skin, the removal process can remove excess amounts of the stratum corneum—the outermost layer of the epidermis—causing subsequent irritation and evoking flare-ups in eczema-prone skin (Matsumura et. al., 2012).

In general, disposable medical device designs are error-prone as they must fulfill numerous stakeholders' requirements while satisfying baseline medical standards. Disposable devices must be affordable, compact, easy-to-use, and sterile. However, they must also meet specific requirements based on what problem they are designed to resolve. For example, the Tegaderm dressing comes in single-use sizes and is packaged to

minimize user contact with the actual product to maintain cleanliness (Chambers, 2025). The product, in order to serve its purpose of accelerating wound healing, also must consider the optimal dimensions to be produced, as well as how it can reduce the peel strength requirement during dressing removal and maintain moisture retention and elasticity. Overall, the seemingly simple design of the disposable dressing must complete several requirements from a healthcare perspective while maintaining a profit margin from a manufacturing perspective.

## Device Analysis

The final product archeology canvas, reflects the key characteristics currently present in the Tegaderm design as well as future requirements that must be considered to improve the device. A majority of the device's current characteristics are guided by the device's purpose and value, which focus on user convenience and maintenance of the optimal conditions required to aid wound healing. For example, the product composition of primarily polyurethane serves to seal in natural moisture produced within the wound site, allowing for increased tissue regeneration (Morgan, 2022). Relevant stakeholder groups guide the remainder of the device requirements and needs. The respective needs of patients, regulators, healthcare providers, and manufacturers all differ according to how each group interacts with the product. For example, patients, specifically those with eczema, are more concerned with the effectiveness of the product than production costs, which is a greater concern for manufacturers. These stakeholder groups are also united by common needs, such as the overall reliability of the product; while patients and healthcare professionals are more personally affected by the product efficacy as the dressing is integrated within treatment plans, regulators (i.e., FDA) and manufacturers must ensure overall consistency in the dressing to avoid legal retributions. The overall impact of the product identified within the canvas is Tegaderm's role as an affordable alternative to traditional dressings.

The most prominent needs that guided device improvement were the physiological needs of patients and the general public, as well as the usability/treatment needs for medical insurance companies. The physiological needs Tegaderm failed to fulfill revolve around its compatibility with diverse skin types, especially sensitive skin prone to irritability and eczema flare-ups. Similarly, the usability/treatment needs for medical insurance companies focused on the dressing's potential to cause further injury or infection at the wound site (Chambers, 2025). The primary problem identified within the current device design combines these requirements, producing one sole issue to be resolved: minimizing the peel strength required during the removal process of the dressing to reduce unnecessary damage to the skin surrounding the wound site.

## Defining the Problem

The ease of use of this device stems from its ability for healthcare workers to monitor wound sites in sensitive, moist environments while keeping the area sterile. When analyzing and researching the Tegaderm clear wound dressing, there were three major components that contributed to its effectiveness: its material makeup, transparency, and adhesive properties. The Tegaderm is composed of a polyurethane film and an acrylic-based hypoallergenic adhesive, creating a sterile product for the woundsite (Food and Drug Administration, 1997). At the same time, the Tegaderm's semi-porous nature and transparency promote optimal moisture retention within the bandage, thereby decreasing the risk of infection and irritation while simultaneously allowing for consistent inspection of the wound site (3M, n.d.).

With the evaluation of the created Stakeholders Needs Matrix, our team focused on improving patient experience for those who face irritation from the device due to the overarching abundance of psychological, financial, and technical needs of this stakeholder group when compared to the needs of other stakeholders like healthcare workers and manufacturers. To determine the device's needed improvement, research was done on the three major components of the bandage for any possible enhancements. The initial goal was to change the

material makeup of the bandage as it was discovered that the polyurethane's antimicrobial property stems from "silver nanoparticles, zinc oxide nanoparticles, and polymers containing chloride," raising concerns for patients with metal allergies (Cui, 2023). However, as the design process continued, this issue became difficult to address due to the rarity of the allergy and the lack of resources to create a newly refined product composition.

This setback shifted our teams' approach toward creating the problem statement. When reevaluating the stakeholder needs matrix, our team realized that there should be an emphasis on solving pre-existing issues that the stakeholders are facing, rather than looking for potential sources of shortcomings. Looking at past user experiences, a common complaint came from patients with rashes, blisters, and chemical burns (U.S. Food and Drug Administration [FDA], 2017). Eczema patients became the solidified target stakeholder of the problem statement, as these stakeholders commonly experience rashes. When considering potential areas of concern for eczema patients, it was concluded that they are more prone to irritation due to the friction between the bandage and dry skin, as well as irritation from the adhesive peeling off the skin. These factors are then taken into consideration when making the problem statement, in which the finalized statement addresses concerns regarding the discomfort faced during the removal process of the dressing.

### **Final Problem Statement:**

Patients with eczema need a gentle, skin-compatible, and comfortable alternative for traditional adhesives used in clear wound dressing to prevent further skin irritation and wound infection while maintaining the dressing's effectiveness in facilitating wound healing and securing medical devices.

### **User Story:**

Ivy, a 68-year-old grandma, needed to have an IV attached to her arm after losing consciousness due to dehydration. Due to her frail nature, the IV needed to be secured with a transparent wound dressing. However, a few hours after the nurses applied the device to her arm, she began to complain about a sudden itchiness and irritation in the area of skin affected. Since she has eczema, the dry skin on her arm was rubbing against the bandage, which led to the irritation. Later on, when the bandage was removed, parts of her skin were peeled off as well, which led to a burning sensation on her exposed skin.

The situation above showcases the lack of compatibility of the Tegaderm clear dressing with more sensitive skin types, such as Ivy, who has eczema, and its potentially adverse effects on the wound site.

### **Design Inputs and Requirements:**

Requirement: The device shouldn't require such excessive force when removing it as it could remove skin from the outermost layer of the epidermis, further damage the skin site.

Input: The peel strength of bandage should be within the range of 50 g/cm<sup>2</sup> to 100 g/cm<sup>2</sup> (Sharma, 2024).

Requirement: Since eczema patients tend to have drier skin, the bandage needs to have optimal moisture retention to prevent the skin from drying out and causing further irritation.

Input: The device's moisture retention rate is within the WVTR (water vapor transmission rate) of 2,000–2,500 g/m<sup>2</sup>/24 h (Nuutila, 2021).

Requirement: A range of sizes is required for the different needs for various wounds. At the same time, a larger surface area might make it more difficult to manage the moisture retention and removal process of the bandage.

Input: The dimensions of the device is the approximate average of the smallest and largest Tegderm, 8.5cm x 10cm (3M, n.d.).

Requirement: The device should be easy for healthcare workers to use and apply to patients.

Inputs: Device must not increase the time for application by more than 50% of the original time taken to apply the dressing.

Requirement: The device must be able to stretch and conform to irregular angles and different contours of the body while maintaining its usability.

Inputs: The device must have an elasticity factor of 0.5 to 20 Pa (Borbolla-Jiménez et al., 2023).

Requirement: The device must be profitable to produce and sell for manufacturers.

Inputs: Device must not increase in cost by more than \$5 for each bandage when mass produced.

## Solution

Our first concept (Figure 1) involved replacing the clear wound dressing's existing acrylate-based adhesive with a silicone-based alternative. This concept was based on the materials used by existing bandages designed for patients with sensitive skin, which employ a similar compound. With a peel force of approximately 50-100 g/cm<sup>2</sup> (Sharma, 2024), this design perfectly matches the ideal range of force needed to remove the dressing without damaging sensitive skin. Furthermore, as the main components of the clear wound dressing remain largely unchanged, a satisfactory level of moisture retention (Nuutila, 2021), sizing (3M, n.d.), and ease of use are preserved in this design. Although this material change does come with additional costs within the manufacturing process, it was determined not to exceed a \$5.00 USD increase in manufacturing. This was specifically calculated by determining and comparing the cost per square centimeter of both a silicone-based bandage and a Tegaderm clear wound dressing. The silicone-based bandages cost per square centimeter was calculated to be \$0.015 USD/cm<sup>2</sup>, whilst the cost per square centimeter of the original device given was calculated to be approximately \$0.008 USD/cm<sup>2</sup>. This means that there would only be an approximate \$0.007

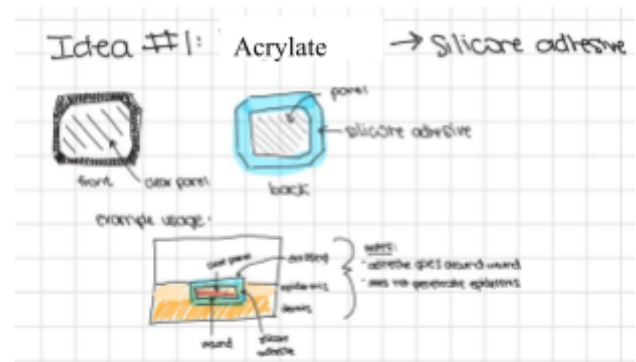


Figure 1. Idea 1; Replacing acrylate with silicone adhesive

USD increase per cm<sup>2</sup> when manufacturing this improved concept, a relatively insignificant cost and effect to the product's profit margins.

Our fifth concept (Figure 2) involved lining the dressing, defined as the area in direct contact with a wound/target site, with a moisturizing compound. Since the adhesive is not altered in this design, the peel force can be assumed to be greater than what would normally be used for patients with sensitive skin, not fully satisfying this requirement. Due to its similar properties,



Figure 2. Idea 5; Line dressing with a moisturizing compound.

this value was assumed to be equal to that of long-wear bandages, which was approximately 250-360 g/cm<sup>2</sup> (Sharma, 2024). Assuming that the moisturizing compound used is similar, if not identical, to the compound used in hydrocolloid dressing, this concept's WVTR can be assumed to be much higher than the necessary transmission rate needed to preserve and enhance wound healing (Nguyen et. al., 2025). This design would also not constrain any other properties, specifically elasticity and ease of use, compared to that of the original

Tegaderm dressing; however, the cost of manufacturing can be predicted to be much greater than that of Idea 1 due to the complexity of hydrocolloid substances. As such, it was determined that this concept was not able to match the need of manufacturability/profitability and thus not considered over the previously discussed concept.

Our sixth concept (Figure 3) involved using a more fluid-reactive adhesive as an alternative. This adhesive would be more sensitive to the patient's sweat/bodily fluids, and when exposed to said compounds, would theoretically loosen and be more easily removed from the patient's skin. This concept would satisfy the requirement for an optimal peel force needed to prevent damage to skin as, prior to removal, this design would not strip any skin it comes in contact with. However, the only caveat is that this would only be the case if the dressing is to be removed when it is properly detached; any instance before contact with sweat/other fluids would have a much greater peel force required, which could, situationally not fulfill this requirement. Additionally, when it comes to keeping wound/target sites moisturized, this design poses a significant problem and does not adequately satisfy this requirement.

Due to the nature of this proposed adhesive, the loosening of the dressing may occur before it is needed to be removed, thus causing the target area of the skin to be exposed to its surrounding environment prematurely. This could be a major detriment to patients who need their wounds to be properly sealed off or medical devices attached to them for a prolonged period of time. This instability would also affect its elasticity as it would reduce the possibilities and angles this dressing could be placed on, as sharp contours would prevent this dressing from staying on the patient for a significant amount of time. In addition to the aforementioned information, this would also not satisfy the requirement for ease of use as this dressing would likely have to be replaced more frequently. The cost for manufacturing of this specific proposal would also be much higher than the previously discussed designs as designing a chemically-reactive adhesive would require extensive amounts of research and additional implementation, thus also not fulfilling the requirement for a relatively profitable and easy to manufacture product.

Among the following concepts, our team's preferred solution was substituting the wound dressing's acrylate adhesive with a silicone alternative (Idea 1). This conclusion was reached mainly through the usage of a weighted decision matrix (refer to the appendix) in which we evaluated all proposed concepts based on how they fit into specific criteria. The criteria, ordered from what was deemed most important to least important, were as follows: peel strength, moisture retention, elasticity, profitability, and ease of use. With this, Idea 1 was determined to be the highest-scoring solution, indicating that it is the most balanced and effective.

To delve into specifics, this solution was chosen as it provided a reliable and clinically proven way to reduce the peel force necessary to remove the dressing from patients, thus reducing the dressing's potential to abrade and harm sensitive skin originating from eczema and other underlying conditions (previously described as the stakeholder's need for the dressing to be gentle). Furthermore, as mentioned before, this concept does not overcompromise on other essential wound dressing features. As the proposal essentially replaces the existing adhesive with an improved alternative, Tegaderm's exceptional moisture retention and elasticity is preserved (3M, n.d.), indicating our product's ability to keep target skin areas healthy and fit along a wide array of bodily contours and angles. The minimal change in the dressing's overall structure also contributes to the proposed device's ease of use, as no additional measures need to be taken in order to apply and secure the dressing (compared to the original device).

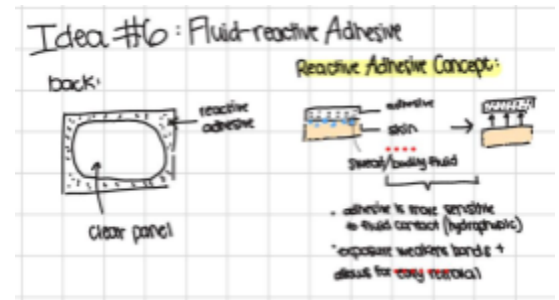


Figure 3. Idea 6; Replacing current adhesive with fluid-reactive alternative

## Prototype

Our team's prototype was a direct sample of a silicone-based adhesive that is commonly found in bandages marketed towards individuals with sensitive skin. This would mimic the adhesive that would be used to replace Tegaderm's current acrylate adhesive (refer to Idea 1). By testing and comparing both kinds of adhesives, our team hoped to ensure that this adhesive would be less abrasive to the skin, through removing fewer layers of the cardboard, than the current adhesive employed by Tegaderm.

This test was carried out by taking an equal-sized (approximately 2cm x 2cm) part of each adhesive and applying it onto a sheet of corrugated cardboard, mimicking human skin. Figure 4 depicts the setup used to test the prototype and compare results to the original product. The surface covered by each type of adhesive is the same for both the silicone and acrylate, and the type of surface simulating skin is the same. The independent



Figure 4. Prototype Testing Set Up.

Silicone adhesive bandages on left and Tegaderm acrylate adhesive on right

variable is the type of adhesive, silicone alternative, or acrylate found in Tegaderm, and the dependent variable is the mass of the cardboard skin sheet that is stripped with the removal of each adhesive device. Both adhesive samples would be removed with approximately the same force. The initial mass of each cardboard sheet, the mass with bandage and clear wound dressing, and the final mass after removal would be recorded and compared to determine the amount of skin stripping that occurred to each cardboard skin sheet due to the removal of the dressings. Prototype testing yielded data supporting that the silicone adhesive resulted in minimal skin stripping and was therefore a more gentle alternative to the acrylate adhesive. The initial, total, and final

masses of the silicone bandage, cardboard skin sheet were: 9.107 grams, 10.857 grams, and 9.100 grams. The initial, total, and final masses of the Tegaderm acrylate adhesive dressing, cardboard skin sheet were: 9.045 grams, 9.564 grams, and 9.024 grams. This shows that the weight of the silicone bandage was 1.75 grams, and removal resulted in the stripping of 0.007 grams of the cardboard skin, while the weight of the Tegaderm clear wound dressing was 0.519 grams, and removal resulted in the stripping of 0.021 grams of the cardboard. This evidence justifies that the silicone adhesive is gentler and less stripping of the skin.

The purpose of conducting this experiment was to compare how the silicone adhesive alternative would differ from the effects of removing the acrylate adhesive. In doing so, the design of our testing allows us to see the effects of our design input and determine if this change positively improves the medical device. This influences our concept by showing that changing the adhesive on the medical wound dressing directly affects how much skin irritation and epidermal stripping occur. Testing and the resulting data showed that our design concept could potentially improve the function of our medical device. We could then compare our results to see if the new adhesive alternative would allow the clear wound dressing to meet all the needs and functions outlined in the problem statement. According to our results, we can determine that a clear wound dressing utilizing a silicone adhesive can be a solution to the problem statement and adequately meet user and stakeholder needs.

## Conclusion

Our design aims to improve the peel strength of the adhesive in the Tegaderm clear wound dressing that was causing skin irritation and epidermal stripping. This caused discomfort and hardship for eczema patients who were more vulnerable to the adhesive's aggressiveness, with removal stripping away at the top layer of the skin. Our new design replaces the acrylate adhesive in the Tegaderm dressing with a silicone adhesive to enhance the suitability and compatibility of clear wound dressings for patients with eczema. The silicone adhesive is gentler and comfortable for the delicate and sensitive skin texture of eczema patients. Our device enhances and adjusts the peel strength of the adhesive for medical wound dressings, allowing for smoother removal that minimizes skin damage.

It can be concluded that our design effectively improves the peel strength of the wound dressings based on the results of prototype testing. The silicone adhesive proved to be three times more gentle in preserving the cardboard simulating the epidermal layer of the skin. This is because the traditional original acrylate adhesive stripped 0.021 grams, while the silicon adhesive only stripped 0.007 grams of the cardboard skin during removal. These results suggest that eczema patients could expect to experience minimal skin stripping when removing a clear wound dressing made with a silicone adhesive, adding to their user comfort and skin compatibility with the device.

The silicone adhesive provides an adhesive for the clear wound dressing that minimizes the stripping of the skin with removal while maintaining the device's ability to protect against moisture, facilitating wound healing while also maintaining visibility of the wound. Prototype testing facilitates the dressing's skin stripping upon removal, demonstrating that the silicone adhesive removed less from the "skin" than the original product, when both are removed using the same force, size of dressing, and same surface modeling human "skin." Potential cons of the solution using the silicone alternative are that the silicone bandage is heavier in weight in comparison to the original product. The bulkiness of a clear dressing with silicone adhesive could reduce the bandage's ability to conform to irregular wound surfaces and result in higher manufacturing costs due to larger packaging requirements. Despite the potential cons in the solution product, the benefits that a silicone adhesive could provide for the wide range of patients with eczema, who would otherwise suffer from harmful epidermal stripping due to the traditional acrylate adhesive, prove that silicone offers a more gentle and skin-compatible design.

Throughout the semester, our team looked at many different potential concepts and designs that we could implement in order to solve the problem statement and meet user needs. Across the semester, while researching potential adhesive alternatives, our team initially looked into a hydrocolloid alternative. However, our team researched further, uncovering that a hydrocolloid adhesive could cause skin irritation due to its extreme stickiness, was more expensive to manufacture, and would not be suitable for testing our prototype. As we continued through the design process, researching and brainstorming more solutions to meet our problem statement, our team also chose to focus on eczema patients' skin stripping instead of allergens. This led our team to decide our design input for a new medical device would be the silicon adhesive clear wound dressing. This was the solution we decided to further explore. We continued the design process, creating a prototype and testing the design to determine if the silicone adhesive could be a potential solution to our problem statement while maintaining device functionality.

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## Team Summary

William Nguyen:

I worked and completed the Solutions portion of the document, including the three sketches/figures. I also worked on the majority of the Prototype section and made minor contributions to the second half of Defining the Problem.

Ivy Nhan

I worked on Defining the problem and the second half of the Executive Summary

Ayushi Patel

Once our team got the materials we needed to have a prototype, I went to a lab and tested our prototype and compared the results to the original product. I then worked to complete the parts of the Prototype and Conclusion sections of the final report.

Tisha Priyadarshi

I completed the following sections of the final design report: Introduction, Device Analysis, and the first half of the Executive Summary.

# Appendix

## Final Product Archeology Canvas

| <b>Product: Clear Wound Dressing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Team members: Ivy Nhan, Tisha Priyadarshi, Bill Nguyen, Ayushi Patel</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <u>Purpose</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <u>Stakeholders</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <u>Requirements</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <u>Impact</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>What does this product do for the user?</p> <p><i>The product is designed for user convenience; from its conformability, which enables the dressing to flex with the patient's skin, to its transparency (including radiological) allows the wound to be monitored continuously while removing the need for the dressing to be removed prior to x-ray scans (Food and Drug Administration, 1997).</i></p> <p>What value does this product create?</p> <p><i>The product helps maintain sanitary conditions, fulfilling the user's health needs by reducing risk of infection at the wound site. The value this product creates is attributed primarily to its ability to provide the user with convenience, regulate their wound site, and remain a cost-effective option while maintaining its reliability as a safe and</i></p> | <p>Who are <u>all</u> the stakeholder groups?</p> <p><i>Patients</i><br/> <i>Patients with sensitive skin (allergic to acrylate adhesive or polyurethane film)</i></p> <ul style="list-style-type: none"> <li>- Regulators</li> <li>- Healthcare providers</li> <li>- Manufacturers</li> </ul> <p>What group traits are relevant to this product?</p> <ul style="list-style-type: none"> <li>- Affordability (of product)</li> <li>- Safety/reliability</li> <li>- Effectiveness</li> <li>- Production cost</li> </ul> | <p>How does the design meet the needs you identified in the previous column?</p> <p><i>The clear, flexible design meets the needs of this device by allowing a wound site to stay covered in a moist environment, which is ideal for the healing process, while also allowing for access to IV and other sites and observation. Being able to monitor the healing process without removing the dressing makes this design meet its needs.</i></p> <p><i>Function:</i> Key functions of a clear wound dressing include creating reduced moisture environment for wounds to heal, protecting an IV, catheter, or central line sites, and allowing for the wound to be visible while covered.</p> <p><i>Form:</i> The form is made of a transparent wound dressing made of flexible polyurethane film. This allows the dressing to cover wounds and IV sites easily and still allows for observation without removal of the covering.</p> | <p>What are the impacts of this design?</p> <p><i>Design of the clear wound dressing minimizes the need for unscheduled dressing changes, which offers user convenience when dealing with hard to handle wound sites, poor bodily condition (feverish or elderly), or staying in high-moisture environments. The dressing also claims enhanced wound healing by sealing in natural wound fluid to maintain a moist environment within the seal, preventing dehydration of the wound bed, which would otherwise occur in standard dry dressings (3M, n.d.).</i></p> <p><i>However, some adverse impacts based on past product complaints include potential for allergic reactions on the skin, ranging from rashes to chemical burns. Furthermore, the adhesive of the dressing provides potential to trap unwanted objects such as hair or small debris particles (U.S. Food and Drug Administration [FDA], 2017).</i></p> |

| <i>easy-to-use sterile, disposable medical device (Food and Drug Administration, 1997).</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Manufacturing:</i> Manufacturing companies use the method of rotary die cutting in the manufacturing process of clear wound dressings. (House, 2023)<br>This process allows for the precise cutting of the plastic medical film into the shapes and sizes of these wound dressings. These sizes and shapes are based on the area of the wound they cover, and the shape depends on the location. Using this technique, manufacturers create wound dressings that have the proper shape to fit around wounds. Based on the application type of the dressing, the manufacturing of the adhesive side changes. This is determined by which method is the most sterile technique. Finally, these products are then packaged individually to ensure they are sterile and can effectively guard against infection on open wounds (Chambers, 2025). In attempts to minimize potential for allergic reactions to the product, the manufacturing process could be altered to produce different shapes and sizes of the dressing with perforations to address uniquely shaped wound sites. | <i>Overall, impacts include offering users an enhanced wound dressing alternative at an affordable price.</i> |
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| <b><u>Characteristics</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b><u>Needs</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                               |
| <p>What things about the product can you observe?</p> <p>What is made of?<br/>-Polyurethane film<br/>-Adhesion part is hypoallergenic, acrylate adhesive (U.S. Food and Drug Administration, 1997).</p> <p>What can it do?<br/>The clear wound dressing protects exposed areas, such as wounds, eyes, and IV sites, while still allowing for the area to be visible for monitoring. They are particularly useful in reducing infection risks due to their protection against moisture and bacteria while still allowing oxygen to pass through the bandage (U.S. Food and Drug Administration, 1997).</p> <p>How do you interact with it?...</p> <p>When interacting with the clear wound dressing, it is important</p> | <p>For each stakeholder group/trait, what needs does the product design address?</p> <p><i>Affordability (for Patients): Price of an application or replacement in hospital settings on a patient can be from \$50 to \$100 (North Coast Medical, n.d.). Outside of hospital settings, these bandages are more affordable with about 20-50 pieces for \$20 on Amazon based on the brand used.</i></p> <p><i>Safety/reliability (for Healthcare providers, Patients, and Regulators): Disposable bandages reduce spread of bacteria. The bandage itself is made from hypoallergenic material that is latex free, which reduces risk of bacteria while being gentle on the skin (3M, n.d.).</i></p> <p><i>Reliability (for Healthcare providers, Patients, and Regulators): The reason for polyurethane's anti-microbial properties is due to its composition of "silver nanoparticles, zinc oxide nanoparticles, and polymers containing chloride" within it (Cui, 2023). At the same time, the polyurethane film provides elasticity and puncture</i></p> | <p><i>Life Cycle:</i> The life cycle of wound dressing starts when it is manufactured. First, the manufacturer must create and develop the products in</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                               |

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| <p>to maintain a sterile technique. Healthcare providers apply the wound dressing by preparing the wound and then removing the adhesive cover. Next then they will center the clear dressing over the wound and apply light pressure to secure the dressing in place, sealing it in place. When positioning the dressing, it is important to identify any nearby regions of skin that may be susceptible to irritation (blisters, open lacerations, etc.) and avoid placing the adhesive directly over the site. Medical professionals must be timely with the wound dressing changes. Manufactures determined based on injury reports regarding infection under the covering are due to not changing the dressing frequently enough and not monitoring the wounds properly. (U.S. Food and Drug Administration [FDA], 2023)</p> <p>Patients interact with their clear wound dressing by living with it daily. A clear wound dressing is waterproof so that the patient can go about their day and take showers without</p> | <p>protection due to the material's high tensile strength and elastic module, which allows it to be flexible (Cui, 2023)</p> <p>Skin compatibility/Sensitive Skin (for different patient skin types and ages): The clear wound coverings need to be designed so that they are safe to use on all skin types. This can mean lighter adhesives for elderly patients with loose skin or babies with highly sensitive skin. There have been reported incidents of Medical Adhesive-Related Skin Injuries to the FDA Maude database. This is an important need to be met to ensure the safety of using the clear wound dressing. (U.S. Food and Drug Administration [FDA], 2020).</p> <p>Effectiveness (for Patients): Known to be effective for patients and areas that are moisture prone. The clear bandage allows for airflow and prevents excess moisture in the area, thus reducing bacteria. At the same time, clear material allows for a visual on the wound area, which is useful for monitoring IV sites (U.S. Food and Drug Administration, 1997). The effectiveness in reducing moisture is important for the clear wound dressing to reduce infections and the formation of blisters. There</p> | <p>a sterile environment. Next, the products must be shipped to a medical institution, like hospitals, clinics, and doctors' offices. All of the transportation, manufacturing, and storage of these products must be in a sterile field to ensure that these products are safe for patient use. Once the product is used and applied to a patient's wound site, the dressing can be worn normally and showered with one. The doctors, patients, and healthcare workers will monitor the dressing, checking if the adhesive becomes loose when it is time to replace the dressing. At that time the product is removed and discarded as medical waste. These products have a one use life cycle.</p> <p>Regulatory: The regulatory needs primarily depend on the safety of the device in terms of its reliability and its clarity in communicating the extent of the dressing's efficacy. For example, the FDA clarifies the restrictions in marketing Tegaderm dressing as synthetic or artificial skin or labeling the device as a cure for any type of wound. This device however does fulfill certain needs for regulators to approve of the device by providing a sterile and hygienic environment for users' wounds to heal.</p> |  |
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| <p><i>having to change the dressing. In the hospital setting, the clear wound dressing is used to cover IVs and catheters. This means that the wound dressing covers these access points, but healthcare workers can still access them. However, venipuncture, blood draws, and injections are not administered at these sites. (U.S. Food and Drug Administration, 1997)</i></p> | <p><i>have been common issues with this, and several reports of blisters forming under the clear film despite the excess moisture prevention design elements. (U.S. Food and Drug Administration [FDA], 2017)</i></p> <p><i>Production Costs (for Manufacturers): Low costs to produce goods (roughly a couple cents a piece) due to mass production strategies. Utilizing rotary die-cutting technology, machines are able to produce high volumes of bandages due to its versatility in combining steps of production while being scaleable (Chambers, 2025).</i></p> |  |  |
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# Stakeholder Needs Matrix

| Stakeholder          | Physiological                                                                                                                                                                                                                                                                       | Financial                                                                                | Usability/<br>Treatment                                                                                                        | Other                                                        |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Patient              | <p>Dressing is flexible (compatible with movement)</p> <p><b>Attaches/sticks and equally effective on multiple skin types</b></p> <p>Safe for skin use, won't cause tearing to the skin, irritation</p> <p>Creates a seal, prevents external moisture from breaching wound site</p> | Cheap and able to be disposable                                                          | <p>Easy to use</p> <p><b>Doesn't stick on itself (or fall off within 2 days)</b></p> <p>Clear/wound is able to be observed</p> | Inconspicuous for aesthetic purposes and patient preferences |
| Patient's loved ones | X                                                                                                                                                                                                                                                                                   | Affordable for guardian/caretaker                                                        | Easy to use in cases where they are responsible for maintaining wound care for patient                                         | Emotional distress if the patient is in distress.            |
| Nurse                | Convenient and easy-to-remove packaging                                                                                                                                                                                                                                             | X                                                                                        | <p>Easy access, easy removal, easy application</p> <p>Sterile technique can be applied</p>                                     | X                                                            |
| Doctor               | Convenient and easy-to-remove packaging                                                                                                                                                                                                                                             | X                                                                                        | Clear appearance enables doctors to easily monitor wound site                                                                  | X                                                            |
| Hospital             | X                                                                                                                                                                                                                                                                                   | Profit margin (cost of purchasing dressings is outweighed by cost of treatments offered) | Standard of care, and standard techniques, that can allow for easy training.                                                   | X                                                            |
| Medical Insurance    | X                                                                                                                                                                                                                                                                                   | Cost of product + fee for treatment →                                                    | The application process and use of                                                                                             | X                                                            |

|                |                                           |                                                                       |                                                                                                                                                                            |                                               |
|----------------|-------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
|                |                                           | how much insurance needs to cover                                     | this device <b>doesn't further injure or directly cause infection (in cases with specific skin types)</b> , and won't leave the hospital, doctor, nurse liable for injury. |                                               |
| Manufacturer   | X                                         | Can be manufactured in bulk, relatively cheap material, profit margin | X                                                                                                                                                                          | Easily distributable                          |
| General public | <b>Compatible with diverse skin types</b> | Affordable, comes in batches                                          | Instructions are clear, everybody can use, self-replaceable                                                                                                                | Easily accessible from common stores/pharmacy |

## List of Proposed Concepts

1. Replacing acrylate adhesive with silicone adhesive
2. Changing the acrylate composition of the entire dressing to a hydrocolloid material
3. Supplemental product, such as a solvent to dissolve the adhesive when removing the dressing (similar to the concept of nail polish remover)
4. Priming product applied to skin before application of dressing—should ideally help reinforce adhesive's strength without increasing peel strength (would allow us to reduce peel strength of dressing)
5. Line dressing with a moisturizing compound to minimize irritation
6. Manufacturing a roll of dressing material with perforated shapes, allowing users to decide what shape bandage to use to completely avoid dressing contact with irritated skin.
7. Replace the adhesive that, over time, reacts to the skin's sweat or body fluid to naturally release the bandage so that it is easier to peel off.
8. Using a wound dressing with a nanofiber material that allows for breathability and less rough contact on the skin
9. Decrease the surface area where the adhesive is applied. This means that the dressing only has adhesive in certain spots that allow for maximum adhesiveness while not sticking all across the cover. (corners have adhesive)
10. Rided textured surface on the clear wound bandage; these patterns reduce the contact surface area to the skin for the clear wound bandage while maintaining its adhesive capabilities.

## Weighted Decision Matrix

|         | Criteria           |               |            |             |               | Results           |                |                    |
|---------|--------------------|---------------|------------|-------------|---------------|-------------------|----------------|--------------------|
| Weights | 28%                | 28%           | 22%        | 5%          | 17%           |                   |                |                    |
| Options | Moisture Retention | Peel Strength | Elasticity | Ease of Use | Profitability | Base Score (X/50) | Weighted Score | Rank of Importance |
| Idea 1  | 8                  | 10            | 7          | 10          | 5             | 40                | 7.93           | 1                  |
| Idea 2  | 6                  | 7             | 6          | 10          | 4             | 33                | 6.14           | 7                  |
| Idea 3  | 7                  | 9             | 7          | 6           | 3             | 32                | 6.76           | 4                  |
| Idea 4  | 5                  | 8             | 7          | 6           | 3             | 29                | 5.99           | 8                  |
| Idea 5  | 10                 | 8             | 7          | 10          | 4             | 39                | 7.76           | 2                  |
| Idea 6  | 7                  | 7             | 6          | 7           | 4             | 31                | 6.27           | 6                  |
| Idea 7  | 5                  | 10            | 7          | 10          | 4             | 36                | 6.92           | 3                  |
| Idea 8  | 5                  | 7             | 5          | 9           | 6             | 32                | 5.93           | 9                  |
| Idea 9  | 7                  | 5             | 6          | 9           | 3             | 30                | 5.64           | 10                 |
| Idea 10 | 7                  | 8             | 7          | 9           | 3             | 34                | 6.70           | 5                  |